MILITARY SPECIFICATION

REFRIGERATION UNITS, MECHANICAL, PANEL TYPE, GASOLINE ENGINE AND ELECTRIC MOTOR DRIVEN, FOR SEMITRAILER REFRIGERATORS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of self-contained refrigeration units for semitrailer refrigerators.

1.2 Classification. The refrigeration units covered by this specification shall be of the following types, as specified (see 6.2).

Type I	-	Refrigeration Unit, Gasoline Engine Driven
Type II	-	Refrigeration Unit, Gasoline Engine and Electric Motor Driven
Type III	-	Refrigeration Unit/Heating Unit, Gasoline Engine Driven

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-F-506	-	Flux, Soldering; Paste and Liquid
P-D-680	-	Dry-Cleaning Solvent
BB-F-1421	-	Fluorocarbon Refrigerants
CC-M-641	-	Motor, Alternating Current (Integral Horsepower)
QQ-A-250/2	-	Aluminum Alloy, 3003, Plate and Sheet
QQ-A-250/8	-	Aluminum Alloy, 5052, Plate and Sheet

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QQ-A-596	-	Aluminum Alloy Permanent and Semi-Permanent Mold Casting
QQ-A-601	-	Aluminum Alloy Sand Castings
QQ-B-654	-	Brazing Alloys
QQ-S-571	-	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy
QQ-S-624	-	Steel Bars, Alloy, Cold Finished and Hot Rolled (General Purpose)
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon
QQ-S-700	-	Steel, Sheet and Strip, Medium and High Carbon
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates and Bars
QQ-T-830	-	Tube, Steel, Carbon, Mechanical, Round, Seamless and Welded
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous, Surfaces For Organic Coatings
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-664	-	Primer, Coating, Synthetic, Rust Inhibiting, Lacquer Resisting
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
VV-G-1690	-	Gasoline, Automotive, Low Leaded or Unleaded
VV-L-825	-	Lubricating Oil, Refrigerant Compressor
WW-T-799	-	Tube, Copper, Seamless (for Use With Solder-Type or Flared-Tube Fittings)

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MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation-Packaging, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture and Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514	-	Plates Identification, Instruction and Marking Blank
MIL-P-1757	-	Primer, Coating, Zinc Chromate, Low Moisture, Sensitivity
MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Heavy Duty
MIL-F-3541	-	Fittings, Lubrication
MIL-W-5086	-	Wire, Electirc, Hookup and Interconnecting, Polyvinyl Chloride, Insulated, Copper or Copper Conductors
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-6872	-	Soldering Process General Specification for

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MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-E-10062 - Engines: Preparation for Shipment and Storage of
MIL-C-10578 - Corrosion Removing and Metal Compound (Phosphoric Acid Base)
MIL-S-12514 - Starter, Motor, Across-The-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 Horsepower (h.p.) Manual and Magnetic
MIL-C-14370 - Compressor, Reciprocating Power Driven Open Type for Use with Dichlorodifluoromethane (CCL_2F_2)
MIL-C-23122 - Condensers, Air-Cooled, Refrigerant 12
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating
MIL-T-46786 - Tanks, Fuel, Engine, Requirements For
MIL-S-62034 - Semitrailer, Van and Semitrailer, Refrigerator, Military Design

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-279 - Compressor, Reciprocating; Power Driven Open Type, For Use with Refrigerant 12
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-759 - Condenser, Air Cooled, Refrigerant 12
MIL-STD-1400 - Engine, Gasoline or Diesel, Methods of Test
MS 16993 - Plug, Fusible, Refrigerant 12
MS 17241 - Heat Interchanger, Refrigerant 12
MS 17243 - Valves, Angle, Shutoff, Packed Receiver, Refrigerant 12
MS 17291 - Receivers, Liquid, Refrigerant 12
MS 17798 - Indicator, Sight, Liquid, Refrigerant
MS 17810 - Valves, Check, Refrigerant 12
MS 17846 - Valves, Pressure Regulating, Crankcase Refrigerant 12
MS 17982 - Valve, Expansion Thermostatic, Refrigerant

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MS 27211 - Thermometer, Indicating Capillary Tube & Bulb
MS 27900 - Gages, Compound, Pressure & Vacuum, Dial Indicating Panel mounted, Refrigerant 12
MS 25068 - Switch, Toggle, Four Pole, Sealed Toggle
MS 25098 - Switch Toggle, One Pole, Sealed Toggle
MS 25273 - Relay, 10 Amp., 2 Pdt. Class B8 Sealed (Asy)
MS 27901 - Gage Pressure, Dial Indicating, Panel Mounted, Refrigerant 12
MS 35845 - Dehydrator, Dessicant, Refrigerant
MS 35867 - Connection End, Internal Flare, Refrigeration
MS 35868 - Connection End, External Pipe Thread, Refrigeration
MS 35869 - Adapter, Straight, Male, Refrigeration Pipe to Flared Tube
MS 35871 - Tees, Pipe to Tube, Refrigeration
MS 35872 - Flare Nut & Flare Seal Bonnet, Refrigeration
MS 35880 - Valves, Packless, Stop, Solder-Joint Connection Ends
MS 35881 - Valves, Packless, Stop, Flare Connection Ends
MS 35882 - Valves, Packless, Stop, Three-Way, Solder Joint Connection Ends
MS 35883 - Valves, Packless, Stop, Three-Way, Flare Connection Ends
MS 35917 - Elbows, Tube, 90 and 45 degree, Solder-Joint Refrigeration
MS 35918 - Adapter, Straight, Pipe Thread to Tube, Male Solder-Joint Refrigeration
MS 35919 - Adapter, Straight, Pipe to Tube, External Flare to Female Solder-Joint, Refrigeration
MS 35920 - Reducer, Tube, Solder-Joint, Refrigeration
MS 35921 - Adapter, Straight, Pipe to Tube; Pipe THD to Female Solder-Joint Refrigeration
MS 35922 - Elbow; Tube, 90 Degree Reducing, Solder-Joint, Refrigeration
MS 35923 - Reducer, Tube, Solder-Joint, Refrigeration
MS 35924 - Plug and Cap, Tube, Solder-Joint, Refrigeration
MS 35925 - Connection End, Solder-Joint Refrigeration
MS 35926 - Connection End, Internal Pipe Thread, Refrigeration
MS 35927 - Coupling, Tube, With Stop, Solder-Joint, Refrigeration
MS 35928 - Elbows, Tube, Street, 90 and 45 Degree, Solder-Joint Refrigeration
MS 35929 - Tee, Tube, Solder-Joint, Refrigeration

QUALIFIED PRODUCTS LISTS

QPL-14370-3 - Compressor, Reciprocating Power Driven, Open Type for use with Dichlorodifluoromethane

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DRAWINGS

ARMY NATICK LABORATORIES

5-13-1957 - Refrigeration Unit, Mechanical, Panel-Type, Gasoline Engine Driven For, Semi-Trailer Refrigerators

(Miniature copy of Drawing 5-13-1957 identified as figure 1 is attached for information purposes only.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

AR-70-38 - Research, Development, Test, and Evaluation for Extreme Climatic Conditions

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Rubber Manufacturers' Association

Standards for Multiple V-Belt Drives

(Application for copies should be addressed to the Rubber Manufacturers' Association, 444 Madison Avenue, New York, NY 10022.)

National Bureau of Standards

Handbook H28 - Screw Thread Standards For Federal Services

(Application for copies of Handbook H28 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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Underwriters' Laboratories, Inc., (UL)

Standard 207C - Refrigerant Containing Components - Feb 1967

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; Walt Whitman Road, Melville, L.I., NY 11749; or 1655 Scott Blvd., Santa Clara, CA 95052.)

United States Department of Commerce

Standard PS-1 - 66 - Softwood Plywood Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American Welding Society

C1.2-53 - Recommended Practices for Spot Welding Aluminum and Aluminum Alloys

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

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* 3.1.1 Initial production testing sample. When specified (see 6.2), the supplier shall furnish a sample for initial production inspection and approval (see 4.4 and 6.4).

* 3.1.2 Initial comparison testing sample. When specified (see 6.2), the supplier shall furnish a sample for comparison inspection and approval (see 4.5 and 6.5).

3.2 Standard component. Each component of the refrigeration units, except Military Standard components, including accessories, instruments and controls, shall be a current standard catalog product of its manufacturer, except for any changes necessary to comply with the specification requirements. No obsolete or discontinued components shall be used.

* 3.3 Code and standards. The refrigerator unit components, as applicable, shall comply with the applicable requirements of UL Standard 207C and electrical components shall comply to UL Standards as applicable.

* 3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that refrigeration unit components he proposes to furnish under this specification meet the applicable requirements of UL Standard 207C and the electrical components he proposes to supply meet UL Standards as applicable.

* 3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the refrigeration unit components have been tested and conform to UL Standard No. 207C and the electrical components conform to the applicable UL Standards.

* 3.4 Materials and components. Materials and components shall be as specified herein (and on the applicable drawings). Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial refrigeration unit provided the completed item complies with all provisions of this specification. Commercial tolerances shall apply to all metal thicknesses and diameters.

3.4.1 Wood.

3.4.1.1 Lumber. All wood used shall be select grade hardwood.

3.4.1.2 Plywood. All plywood shall be group I, exterior type, grade B-C, sanded on two sides and shall conform to PS-I-66.

3.4.1.3 Wood preservative. Wood preservative shall conform to P-D-680 and shall contain 1 percent solubilized copper-8-quinolinolate free of amines and napthenic acid or its derivative.

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3.4.2 Bolts, nuts, screws, and washers. Bolts, nuts, screws, and washers used for mounting components shall be of the quality normally used for the purpose in commercial practice. Threads shall conform to the National Bureau of Standards Handbook H28. Steel bolts, nuts, and washers for the evaporator shall be zinc plated. Steel bolts, nuts, screws, and washers on sections other than the evaporator shall be zinc or cadmium plated. Brass bolts, nuts, washers, and machine screws shall not be used in fastening aluminum parts.

3.4.3 Gasket. The gasket for the heat shield shall be fabricated of a resilient sponge rubber core incased in an oil resistant synthetic rubber skin and in the cross-section dimensions shown on Drawing 5-13-1957. The gasket core may be either natural or synthetic rubber. The minimum thickness of the skin shall be 0.015 inch.

3.4.4. Metal conditioner.

3.4.4.1 Ferrous metal. Ferrous metal conditioner shall be type I or type II of MIL-C-10578 or shall conform to type I or II method VI of TT-C-490.

3.4.4.2 Aluminum metal. Aluminum surface conditioner shall conform to MIL-C-5541.

3.4.5 Primer and paint.

3.4.5.1 Metal primer. Primer used on metal surfaces to be painted shall conform to TT-P-1757.

3.4.5.2 Finish paint. The finish coat of paint for ferrous and aluminum metal shall be semi-gloss enamel conforming to TT-E-529. The color shall conform to number 24087 of FED-STD-595.

* 3.4.6 Refrigerant. The refrigerant shall be type 12 of BB-F-1421.

3.4.7 Oil. Refrigerant compressor lubricating oil shall conform to type II of VV-L-825 or may be oil recommended by compressor manufacturer.

3.4.8 Steel.

3.4.8.1 Sheet and strip. Sheet or strip steel shall conform to QQ-S-698 and QQ-S-700.

3.4.8.2 Bar stock. Bar stock steel shall conform to QQ-S-624.

3.4.8.3 Tubing. Steel tubing shall be welded or seamless conforming to QQ-T-830.

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3.4.8.4 Structural. Structural steel shall conform to QQ-S-741.

3.4.9 Aluminum.

3.4.9.1 Sheet or strip. The aluminum sheet or strip shall be alloy 5052-H32 conforming to QQ-A-250/8 or alloy 3003-H14 conforming to QQ-A-250/2.

3.4.9.2 Castings. The aluminum castings shall be sand or permanent mold fabricated. When sand castings are supplied, they shall conform to alloy 195 or 108 of QQ-A-601. When permanent mold castings are supplied, they shall conform to alloy B195 of QQ-A-596.

3.4.10 Joining materials.

3.4.10.1 Silver for brazing. Silver brazing alloy shall conform to QQ-B-654.

3.4.10.2 Solder. Solder shall conform to composition Sb5 of QQ-S-571.

3.4.10.3 Soldering flux. Soldering flux shall conform to type I of O-F-506.

* 3.4.11 Tube, copper. Tube used for refrigerant lines with an outside diameter of 5/8 inch or less may be hard copper tube conforming to type K or L of WW-T-799 or soft copper tube. The minimum wall thickness of soft copper tube shall be as follows: 1/4 inch tube - 0.030 inch wall thickness; 3/8 and 1/2 inch tube - 0.032 inch wall thickness; 5/8 inch tube - 0.035 inch wall thickness. Refrigerant tube with an outside diameter of 7/8 inch or more shall be type K or L soft or hard copper tube conforming to WW-T-799. Tube used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness not less than 0.016 inch.

3.4.12 Insulation. Insulation shall have a commercial manufacturer's rated K factor of not more than 0.15 British thermal units per hour (Btuh), per square foot per inch of thickness per degree Fahrenheit temperature difference at a mean temperature of 75 degrees Fahrenheit (F.). Insulation shall be odor free and shall be of a fire retardant type.

* 3.5 Design. The type I and II refrigeration units shall be furnished complete with accessories, instruments, and necessary controls to provide for temperature selection between 35 F. and 50 F. by refrigerant by-pass; to provide for zero or below temperature without refrigerant by-pass and to provide defrost. The type III units shall be furnished complete with accessories, instruments, and necessary controls to provide for automatic refrigerator temperature selection between 0 F. and 50 F. and to provide for automatic defrost. The type III units shall also have a heating capability as specified in 3.5.1.1.3. The type I and III units shall be gasoline engine driven only and the type II

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units shall be gasoline engine and electric motor driven. The type II units shall allow for mounting the electric motor on the left side when facing the front of the unit. The net weight of complete refrigeration unit shall not exceed 1200 pounds (lbs) for the type I and III units and 1400 lbs for the type II units. The refrigeration units shall be capable of starting or being started and operating in a condenser inlet air temperature of plus 120 F. (see 4.8.4). The refrigeration units shall be designed to withstand a 3600 mile road endurance test as specified in 4.8.14. The refrigeration units shall be designed to pull down an empty refrigerator, semi-trailer from 110 F. to 0 F. in 8 hours when tested as specified in 4.8.15. The ambient will be maintained at 110 F. during the test. The refrigeration units shall show no signs of failure when impact tested 200,000 times at 5G impact as specified in 4.8.7. The refrigeration units and all components thereof shall be designed to withstand a 6 inch drop without a 1/32 inch distortion or misalignment or 1/8 inch displacement of condenser and evaporator components, refrigerant leaks, or damage when tested as specified in 4.8.8. The refrigeration units shall be designed to operate and continue operating in a 3 inch per hour rainfall test specified in 4.8.6. Refrigeration units shall conform to the design and dimensions shown on Drawing 5-13-1957, except that the motor for the type II may protrude on the left outside the frame. Externally mounted motors shall have a pulley guard on the portion extending beyond the unit inclosure. The refrigeration units shall be designed to operate using the refrigerant specified in 3.4.6. The refrigeration units shall be capable of operating as intended when tested as specified in 4.8.1 and 4.8.2. The completely assembled refrigeration unit shall be capable of withstanding storage temperatures of plus 155 F. and 10 to 20 percent relative humidity for 4 hours and minus 30 F. for 6 hours without any evidence of deterioration, corrosion, damage, such as rupture of tubing, damage to control components, binding of motors, fans, or other mechanical parts, deterioration of gaskets, breaking down of adhesives, permanent setting of gaskets, or loss of refrigerant, when tested as specified in 4.8.13. The refrigeration unit shall operate properly without malfunction, excessive noise or vibration as a result of exposure to sand and dust when tested as specified in 4.8.16.

3.5.1 Performance characteristics.

3.5.1.1 Capacity. The refrigeration units shall meet the capacity requirements specified herein and tested as specified in 4.8.3.

3.5.1.1.1 Cooling capacity at 0 F. refrigerator temperature. The low temperature net capacity for the refrigeration units exclusive of evaporator fan heat gain, shall be not less than 9,000 Btuh with the compressor discharge pressure at no time exceeding 200 pounds per square inch gage (psig). The capacity shall be obtained when the air entering the condenser is at a temperature of 110 F. plus or minus 2 F., the temperature of the air entering the evaporator is 0 F. plus or minus 1 F., the temperature of the air leaving the evaporator is minus 4 F., plus or minus 1 F., and the refrigerant temperature, considered as refrigerant saturation temperature corresponding to the pressure at the evaporator outlet, is not lower than minus 12 F.

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3.5.1.1.2 Cooling capacity at 35 F. refrigerator temperature. The high temperature net capacity for the refrigeration units exclusive of the evaporator fan heat gain, shall be not less than 13,500 Btuh with the compressor discharge pressure at no time exceeding 200 psig. The capacity shall be obtained when the air entering the condenser is at a temperature of 110 F. plus or minus 2 F., the temperature of the air entering the evaporator is 35 F. plus or minus 1 F., and the refrigerant temperature considered as refrigerant saturation temperature corresponding to the pressure at the evaporator outlet is not lower than 12 F.

3.5.1.1.3 Heating capacity, for type III only, at 35 F. refrigerator temperature. The gross heating capacity, including evaporator fan heat gain, shall be not less than 7,000 Btuh. The capacity shall be obtained when the temperature of the air entering the condenser is minus 25 F. plus or minus 1 F., the air entering the evaporator is 35 F. plus or minus 1 F., and the air leaving the evaporator is not higher than 40 F.

* 3.6. Construction. The refrigeration unit shall be fabricated as specified hereinafter. All parts shall be mounted to minimize vibration and shall be corrosion-resistant material or treated to withstand corrosion during field operation. Accessories and piping or tubing shall be fastened and braced to prevent damage or loosening during transit or operation. All component parts of the refrigeration units of a given type furnished under this specification and on any one contract shall be interchangeable. After assembly the refrigeration unit and all components shall function as intended. All refrigerant containing components and piping shall be gas tight when tested as specified in 4.8.10. All wood members used in the fabrication of refrigeration units shall be immersed in wood preservative specified in 3.4.1.3 for a period of not less than 30 minutes and air dried for a minimum of 24 hours. Necessary trimming or resurfacing of completely milled, and treated wood members will be permitted to secure the required fit provided surfaces are thoroughly coated with wood preservative specified herein prior to assembly. The gasoline engine and compressor shall be mounted on a base plate fabricated from steel sheet, nominal thickness 0.1875 inch. Joints (including fasteners) between dissimilar metals in direct contact shall be avoided unless protected against galvanic corrosion. Protection against galvanic corrosion may be accomplished by separation of dissimilar metals by means of zinc chromate coating or plating. All electrical wire shall be in accordance with type II of MIL-W-5086 and shall be arranged and supported to prevent sagging and chafing, and numerically coded for easy identification with the wiring diagram supplied by the manufacturer.

3.6.1 Welding. All welding shall be in accordance with the manufacturer's standard practice for steel and AWS C1.2.53 for spot welding of aluminum and aluminum alloys. The surfaces of parts to be welded shall be clean and free of scale or other foreign matter. All scale or flux deposits shall be removed from the finished welds.

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3.6.2 Soldering and brazing. The soldering or brazing process shall be as described in MIL-S-6872 for soft soldering and MIL-B-7883 for silver brazing.

3.6.3 Refrigeration unit inclosure.

3.6.3.1 Base, framework, and vertical structure. The refrigeration components shall be inclosed in a welded rectangular frame fabricated from aluminum specified in 3.4.9.1, minimum thickness of 0.150 inch or of steel specified in 3.4.8.1 minimum thickness of 0.119 inch. The structural framework shall have a removable bottom plate of aluminum sheet specified in 3.4.9.1 of 0.150 inch minimum thickness, or steel sheet specified in 3.4.8.1 of 0.119 inch minimum thickness. The bottom plate shall protect the gasoline tank from external damage and shall have an access opening so that gasoline may be drained from the gas tank. The power unit base shall be attached to the frame above the gasoline tank. A 1/4 inch reinforcing plate 3-inches by 6-inches shall be welded to the frame to form backing plates for all 6-mounting bolt holes. A vertical heat shield structure shall be secured at the rear of the condensing unit inclosure and at right angles to the power unit base. Two lifting rings of D-ring design constructed to fold down shall be provided and permanently attached to the top of the structural framework. The lifting rings shall be located on the vertical plane through the center of gravity of the refrigeration unit and each ring shall be capable of supporting the entire weight of the unit (see 4.8.11). Hinged access doors shall be provided on the front and a removable access panel shall be provided on the top of the condensing unit enclosure. Access doors and panels shall be constructed of flattened expanded sheet steel having a minimum thickness of 0.089 inch and a sheet steel 1-inch border having a minimum thickness of 0.089 inch and formed to minimize warping or twisting of the assembled door panel. Each access door shall be secured with two barrel bolt latches not less than 2-1/2 inch size as illustrated on Drawing 5-13-1957.

3.6.3.2 Heat shield. The insulated heat shield shall separate and insulate the evaporator section from the condensing unit section. The heat shield shall be constructed with lumber specified in 3.4.1.1 or plywood specified in 3.4.1.2 and braced with wood, steel, or aluminum to support the evaporator section against vibration and shock imposed upon it in service or transportation. The condenser side of the heat shield shall be covered with the same material as the refrigeration unit frame a minimum of 0.0359 inch thick. The evaporator side shall be covered with sheet aluminum specified in 3.4.9.1 minimum thickness of 0.048 inch. All space between sheathing (if used) framing, and metal covering shall be completely filled with a minimum thickness of 3-1/2 inches of insulation specified in 3.4.12. No metal or other high heat conductor shall extend through the insulation except for the fan shaft, damper control, refrigerant piping, condensate drain, and electrical wiring. All holes extending through the heat shield shall be sleeved or grommeted to prevent abrasion between the tubing or wiring and the metal skin. The holes shall be sealed with rubber or asphaltic-base material compounded for use without heating.

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3.6.3.3 Condenser guard. A condenser guard shall be supplied to protect the condenser fins from external damage. The guard shall provide physical protection, while allowing air flow across the condenser coil. The condenser guard shall be fabricated from flattened expanded steel sheet having a minimum thickness of 0.059 inches and treated to resist corrosion. The guard shall have a metal border and be detachable without removing or loosening any other major component.

3.6.4 Evaporator section. The evaporator section shall consist of the evaporator coil, air recirculating fan, expansion valve, and heat interchanger and housing. The construction shall provide for access from inside the refrigerated space to service the fan and the expansion valve. Cadmium which can contact the food stored in the refrigerated space shall not be used in the evaporator section.

3.6.4.1 Evaporator housing. The evaporator housing shall be constructed of aluminum specified in 3.4.9.1. Aluminum shall have a minimum thickness of 0.100 inch. The evaporator housing shall have an air discharge outlet and air intake inlet located for maximum efficiency of cold air circulation that will result in a minimum of air "short circuiting." A drip pan shall be provided at the bottom of the evaporator to collect condensate which shall flow through a drain in the heat shield to a trap located outside the heat shield and discharged outside of the condensing unit enclosure.

3.6.4.2 Evaporator coils.

3.6.4.2.1 Type I and type II units. The evaporator shall have coil capacity to meet the requirements specified in 3.5.1.1.1 and 3.5.1.1.2 and shall utilize hot gas defrosting. The evaporator coils shall be multipass, forced convection, fin type made of copper tubing specified in 3.4.11, expanded into aluminum extended fin surfaces. The fins shall be not less than 0.009 inch thick aluminum sheet and shall be spaced not more than six per inch. The tube assembly shall be either staggered or in line arrangement. The coils shall be equipped with a multiple outlet type refrigerant distributor. The distributor may have a side outlet for a defrost connection or a separate defrost header may be furnished.

3.6.4.2.2 Type III unit. The evaporator shall be a multipass, forced convection fin type made of copper tubing expanded into aluminum extended fin surfaces. Copper tubing shall be as specified in 3.4.11. The fins shall be not less than 0.009 inch thick aluminum sheet and shall be spaced not more than six per inch. Either staggered or in line tube assembly will be acceptable. Coils shall be equipped with a multiple outlet type refrigerant distributor. The evaporator shall be of a design to provide for its performance under reverse-cycle heating conditions as required in section 3.5.1.1.3.

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3.6.4.2.3 Evaporator fan. The evaporator fan shall be a statically balanced propeller type mounted on a shaft extending through the insulated heat shield. The fan hub shall be equipped with two cup point set screws located at 90 degrees to each other. The fan shall be constructed of steel treated to resist corrosion, corrosion-resistant metal, or a combination of these materials. The fan shall be designed to force air through the evaporator with a discharge duct concentrating the air discharge at the top third of the evaporator unit.

3.6.4.2.4 Fan drive. A common shaft shall be used for the condenser and evaporator fans and shall be 1-inch diameter, and shall be step machined with a minimum 1/16-inch radius to accept the fans. Two, 1-inch, pressure lubricated, 4-bolt flange type, ball bearings shall be used to support the fan shaft. A clutch as specified in 3.6.5.6.1 and 3.6.5.6.2 shall be furnished and installed on the fan shaft.

3.6.4.2.5 Expansion valve. The expansion valve shall conform to 17982-4 of MS 17982.

3.6.4.2.6 Heat interchanger. A heat interchanger shall be provided and installed in the evaporator section. The heat interchanger shall conform to 17241-7 of MS 17241.

3.6.5 Condensing section. The condensing section shall consist of prime mover, condenser, compressor, condenser fan, a receiver, dehydrator, and other necessary refrigeration accessories.

3.6.5.1 Compressor. The compressor shall conform to type II of MIL-C-14370 and MIL-STD-279. The compressor shall be installed to be interchangeable on any one contract with all other type II compressors listed in QPL 14370-3. Mounting bolts for the compressor shall be 3/8-16 UNC 2A and the base shall have captive nuts for the removal of the compressor. The compressor shall be installed so that the crankcase oil lever may be observed.

3.6.5.2 Condensers. The inlet and outlet connection of condensers shall be made with solder specified in 3.4.10.2 and the piping shall be installed to permit removal or replacement of the condenser.

3.6.5.2.1 Type I and type II. The condenser for type I and type II refrigeration units shall be an air-cooled type conforming to type I, class I of MIL-STD-759 and MIL-C-23122. The condenser shall be installed so that it can be removed and replaced through the front of the refrigeration unit enclosure.

3.6.5.2.2 Type III. The condenser for type III refrigeration units shall be air-cooled, made of copper tubing and aluminum extended surfaces. Copper tubing shall be as specified in 3.4.11. The aluminum fins shall be not less than 0.009 inch thick, spaced not more than eight per inch. The aluminum fins

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shall be given a chemical film coating conforming to MIL-C-5541. The condenser coil shall have a staggered or in line tube arrangement. The condenser capacity and design shall provide for performance under reverse cycle heating conditions. The airflow over the condenser shall be controlled by manually operated dampers so arranged that when necessary for the heating cycle, the air will be recirculated within the condensing section compartment and over the air cooled engine as a heat source. The condenser shall be installed so that it can be removed and replaced through the front of the refrigerator unit enclosure.

3.6.5.2.3 Condenser fan. The condenser fan shall conform to type I of MIL-STD-759. The condenser fan shall be statically balanced. The fan shall draw air through the condenser coil and discharge it through the top and sides of the refrigeration unit enclosure. The fan hub shall be equipped with two cup point set screws located at 90 degrees to each other.

3.6.5.3 Receiver. The receiver shall conform to MS 17291-3 of MS 17291. The receiver shall be fitted with a fusible plug conforming to 16993-2 of MS 16993 and inlet and outlet valves conforming to MS 17243 sized to fit refrigerant lines shall be installed in the receiver.

3.6.5.4 Dehydrator. The dehydrator shall conform to 35845-3 of MS 35845. A three-valve by-pass shall be installed in the refrigerant line to facilitate replacement of the dehydrator and the receiver outlet valve shall be of the three valves.

3.6.5.5 Liquid moisture and sight indicator. A liquid moisture and sight indicator shall be installed in the liquid refrigerant line conforming to 17798-11 of MS 17798. The indicator shall be installed so that it is possible to observe refrigerant condition from the front of the refrigeration unit.

3.6.5.6 Defrost system.

3.6.5.6.1 Type I and type II units. The type I and type II units shall be equipped with a manually controlled hot gas defrost system. The system shall include a hand shut-off service valve, a solenoid valve in the hot gas line, an electrically controlled fan clutch and a drain pan with hot gas heating coil. The electrically controlled fan clutch shall be used to stop the fans during defrost. The solenoid valve and clutch shall be controlled from the panel by a switch conforming to 25098-23 of MS 25098. Defrost completion shall be indicated by a red pilot light activated by a thermostat located on the suction line between the evaporator coil and interchanger. The thermostat setting shall be set by the manufacturer. When the red pilot comes on it shall indicate the coil is free of frost. The defrost system shall be capable of operating through a complete cycle without overloading the engine by stalling or exceeding the nameplate amperage of the motor. The defrost system shall reclaim 10 pounds of condensate within 30 minutes when tested as specified in 4.8.5.

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3.6.5.6.2 Type III unit. For defrosting, the reverse cycle system shall be utilized using the condenser as an evaporator and conversely the evaporator as a condenser. The defrost timer will periodically actuate the four-way valve to convert the system from refrigeration cycle to heating cycle. The system shall include an automatically controlled evaporator fan drive clutch or a fan damper to prevent air circulation during the defrost. Defrosting action may be terminated by a metallic element, contact thermostat. The defrost system shall reclaim not less than 10 pounds of condensate within 30 minutes (see 4.8.5).

3.6.5.7 Refrigeration control system.

3.6.5.7.1 Temperature control, type I and type II units. The control system for the type I and type II units shall be accomplished by means of a thermostatic hot gas by-pass control valve to provide for the maintenance of a refrigerated space temperature in the range of plus 35 F. to plus 50 F., with a tolerance of plus or minus 3 F., of any preselected temperature within the stated range. A discharge pressure regulating valve shall be provided in the discharge line between the hot gas defrost line and the condenser inlet. This valve shall maintain the discharge pressure from the compressor at a minimum 150 psig during normal operation. This valve shall be equipped with a damping device so it will withstand the pulsation of the discharge gas. Low temperature control shall be obtained by use of a hand shut-off valve isolating the high temperature control system and by manually starting and stopping the unit, as required.

3.6.5.7.2 Temperature control, type III units. Temperature control of both heating and cooling shall be accomplished by automatic starting and stopping the power unit, as required to maintain the desired temperature through the use of a thermostat. A manual switch, selected from and conforming to MS 25068, shall allow the choice of heating or cooling operation by means of a four-way reverse cycle system. In extremely low ambients the louvers of the unit enclosure are to be manually closed to conserve and utilize the heat given off by the gasoline engine. The system shall be complete with any necessary relays conforming to MS 25273, toggle switches conforming to MS 25068 or MS 25098, safety controls, cranking limit switch, automatic choke, and other necessary devices. All controls shall be visible and serviceable from the front of the units.

3.6.5.8 Thermometer. A dial thermometer conforming to 27211-2 of MS 27211 shall be supplied with the remote bulb secured in the return airstream on the inlet side of the evaporator to indicate the refrigerator temperature. The thermometer shall be mounted on the front of the unit as indicated on Drawing 5-13-1957.

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3.6.5.9 Safety cutout switch. A high pressure safety cutout switch with manual reset on type III and automatic reset for type I and II, shall be provided to stop the power unit when the compressor discharge pressure exceeds 250 psig when tested as specified in 4.8.1. For the type I and II units the high pressure cutout switch shall stop the unit by de-energizing the control circuit relay. A red light, mounted to be visible from the front of the unit, shall be wired to light when the power unit stops due to compressor discharge pressure exceeding 250 psig. For types I and II refrigeration units, the red light may be the same light specified in 3.6.5.6.1.

3.6.5.10 Gages, compound and pressure. The compound and pressure gages shall conform to 27901-1 of MS 27900 and 27901 of MS 27901 respectively except that 1/4-inch flare connections shall be acceptable. Gages shall be properly identified and shall be mounted on the front of the condensing section inclosure and connected to the back seating connections of the compressor shut-off valves.

3.6.5.11 Shut-off line valves. The shut-off line valve for refrigeration lines shall conform to MS 35880.

3.6.5.12 Crankcase suction pressure regulating valve. An adjustable crankcase suction pressure regulating valve shall be mounted in the refrigerant suction line between the compressor and the suction accumulator. The valve shall conform to 17846-9 of MS 17846 except that the valve shall be a long connection type and shall be installed to provide a 6-inch clearance for adjustment. The crankcase suction pressure regulating valve shall be adjusted to limit suction pressure at the compressor to 10 psi, maximum.

3.6.5.13 Reverse cycle expansion valve. An expansion valve conforming to MS 17982 shall be provided for controlling the flow of liquid refrigerant during reverse cycle operation on type III units.

3.6.5.14 Four-way valve. On type III units, a four-way reversing valve shall be provided for reverse cycle operation.

3.6.5.15 Check valves. Check valves conforming to MS 17810 shall be installed wherever required for proper starting, operation, and defrosting of the unit.

3.6.5.16 Piping. All tubing used shall conform to the material specified in 3.4.11 except gage and control connecting lines shall be high pressure flexible hose. Minimum sizes of copper tubing with related valves and fittings shall be as follows: 1-1/8 inch outside diameter (O.D.) suction line, 1/2 inch O.D. liquid line, and 7/8 O.D. discharge and 5/8 inch O.D. hot gas defrost line and liquid drain line from condenser to receiver. Valves and fittings shall conform to MS 16993, MS 35867 through MS 35869, MS 35871, MS 35872, and MS 35880 through MS 35883, and MS 35917 through 35929, as applicable. Connections shall

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be solder-joint type, except that flared connections shall be used on the expansion valve, dehydrator, gages, by-pass and receiver valves. The condenser piping shall have flanged connections (see 3.6.5.2). Tubing shall be arranged to resist vibration. Particular attention shall be given to compressor connections. Vibration eliminators or flex tubing shall be installed in piping connections to the compressor. The flexible length of the vibration eliminators shall be not less than 7 inches. All solder connections shall be made with solder specified in 3.4.10.2, except that brazing may be used whenever brazing temperatures will not be deleterious to any fitting or component adjacent to the brazed joint. All flux and foreign material shall be removed from finished joints and tubing and fittings shall be given a coat of protective lacquer after successfully passing the leak test. All components to be soldered or brazed having internal parts that could be damaged during this process, shall be disassembled before soldering or brazing.

3.6.5.17 Suction accumulator. A vertical suction accumulator, nominal 6 inches in diameter and 12 inches high with the inside diameter connections to accommodate O.D. 1-1/8 inches, shall be installed in the refrigerant suction line between the compressor and evaporator to minimize the entrance of liquid refrigerant into the compressor crankcase. Provisions shall be made for return of entrained lubricating oil to the crankcase.

3.6.6 Condensing unit prime movers.

3.6.6.1 Gasoline engine. The gasoline engine shall be multicylinder, air cooled, 4-cycle, capable of producing not less than 7.1 brake horsepower (bhp) at a maximum speed of 1,800 revolutions per minute (rpm), under continuous operation and shall be the engine manufacturers standard commercial engine. The engine shall be capable of operating on lube oil conforming to MIL-L-2104 and gasoline conforming to VV-G-1690. The engine shall be provided with a 12 volt electrical cranking system with provisions for a manual cranking system, either rope or hand crank, and shall be complete with electric starting motor, generator or alternator and all necessary circuit devices and controls and regulators, and mounting provisions. All serviceable components of the engine such as air cleaner, spark plugs, oil dip sticks, oil filler tube, oil drain, fuel filter, and carburetor shall be accessible from the front of the units for service in the engine compartment without removal of the engine or any other components of the refrigeration unit. The crankcase drain hose shall have the same inside diameter (ID) size as the drain opening in the crankcase. Battery, battery cables, and battery compartment will be furnished by the semi-trailer manufacturer as components of the semi-trailer. The air cleaner shall be a heavy duty, oil bath type. The oil pan shall have a minimum oil capacity of 3 quarts. Each production gasoline engine shall be tested as specified in 4.6.17, except those selected for production control test.

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3.6.6.1.1 Clutch. A ball bearing type centrifugal clutch designed to provide for "no load" starting and continuous override operation shall be furnished by the supplier and mounted on the drive of the engine. The clutch shall have a minimum 10 HP rating at 2000 rpm. The clutch shall engage at a speed recommended by the manufacturer after reviewing the operating speeds and control cycle.

3.6.6.2 Fuel tank. A 12-gallon capacity fuel tank in accordance with type I or III of MIL-T-46786 shall be furnished. The fuel tank shall be installed so that the gas tank will be supported by an independant "U" type suspension bracket attached to any part of the frame except the base portion. Construction and mounting shall be such that neither vibration, nor excessive tightening of mounting nor contact with other members of the engine or end item, will cause leaks to develop. A chain-secured type filler fuel tank cap, a filler neck, and a strainer with a clear opening of not less than 2 inches in diameter, all in conformance with item 1, table II of MIL-T-46786 shall be provided. The removable strainer shall be located or protected to prevent contact with the filling nozzle. The filler neck shall be provided with an air vent to prevent "blow back" on rapid filling and shall be located as shown on Drawing No. 5-13-1957. A drain fitting to enable draining without removing the tank from the unit shall be provided. The entrance of the fuel supply pickup tube shall be not more than 1/4 inch above the bottom of the fuel tank. The gas tank filler neck shall extend outiside the condensing section inclosure to allow for filling the gas tank.

3.6.6.3 Controls. All controls including temperature control valve shall be accessible from the front of the unit, and all indicating controls, gages, and meters shall be visible from the front of the unit and shall be located in the approximate location indicated on Drawing 5-13-1957. The control circuit for the type II shall be 12 volts (dc). The supplier shall furnish a transformer rectifier to supply voltage when the electric motor is used as the prime mover.

3.6.6.3.1 Starting controls for type I. For the type I units the engine starting controls shall consist of an "ON-OFF" switch conforming to 25098-22 of MS 25098, a momentary contact starter switch conforming to 25098-30 of MS 25098, and a commercial starter relay. The starter switch shall energize the relay when the switch is in the "ON" position. A remote type manual choke control shall be furnished and shall be operable from the front of the unit.

3.6.6.3.2 Starting controls for type II. For the type II units the starting controls shall consist of a double throw selector switch conforming to 25068-21 of MS 25068 for selection of electric motor or gasoline engine operation. The switch shall be marked "OFF" on the center position, "electric motor" on the second position, and "gasoline engine" on the third position. A momentary switch conforming to 25098-30 of MS 25098 shall be supplied for energizing the starter solenoid. The switch shall be marked "start" on the momentary position and "run"

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on the other position. For the gasoline engine control circuit a relay conforming to MS 25273 shall be furnished. When the relay is de-energized it shall ground the engine magneto and open the circuit to the clutch and defrost solenoid. A remote type manual choke control shall be furnished and shall be operable from the front of the unit.

3.6.6.3.3 Starting controls for type III. For the type III units the engine starting controls shall consist of an "ON-OFF" starter switch conforming to 25068-24 of MS 25068, an overcranking circuit breaker switch, an automatic choke control, and necessary starter relays conforming to MS 25273.

3.6.6.3.3.1 Overcranking circuit breaker. The control circuit shall contain an overcranking circuit breaker so designed that if the engine fails to start within 30 seconds plus or minus 10 seconds after cranking begins, the starting circuit will be broken (see 4.8.1). The circuit breaker shall contain a temperature and current responsive element which shall serve as a safety device and shall be reset manually. The circuit breaker shall have either color or position indication to show when the starting circuit is broken.

3.6.6.3.3.2 Automatic electric choke. The supplier shall furnish an automatic electric choke mechanism designed and installed to automatically control the opening and closing of the carburetor choke to enable the engine to start and terminate the "choking" as consistent with engine performance.

3.6.6.3.4 Gasoline gage. A 1-1/2 inch diameter iron pipe thread gasoline gage shall be supplied consisting of a liquid level indicator with mechanical connections to a float and linkage mechanism in the fuel tank. The gasoline gage shall be horizontally installed in a manner that will be visible and will permit removal and replacement from the front of the unit.

3.6.6.3.5 Engine hour meter. A nonresetting engine hour meter for recording engine operation time shall be provided. The meter shall be of the type and voltage for the application and use with the refrigeration unit on which installed, and shall be capable of recording a total of 9,999 hours. The mounting shall be done to prevent damage by operational vibrations. The meter shall be installed to permit removal and replacement from the front of the unit on all types.

* 3.6.6.3.6 Ammeter. A 2-1/16 inch flange and flush mounted automotive type ammeter with a clear plastic crystal shall be supplied. The ammeter shall be installed to permit removal and replacement from the front of the unit.

* 3.6.6.3.7 Oil pressure gage. A 2-1/16 inch case and flush mounted automotive type oil pressure gage shall be supplied and shall be calibrated over the range of 0 to 50 psi. The oil pressure gage shall be mounted in a visible location and installed to permit removal and replacement from the front of the unit on all types.

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3.6.6.4 Electric motor for type II units. A 7-1/2 HP, 1750 rpm electric motor shall be provided for the type II unit to operate from a nominal 208-volt, 3-phase, 60-Hertz power supply. The motor shall be of the squirrel-cage type, NEMA Design "C", with a NEMA 213T frame, designed for high torque starting, continuous duty, protected by a drip-proof enclosure and shall conform to CC-M-641.

3.6.6.5 Motor starter for type II units. A general purpose motor starter conforming to type I, class 2, size 1 of MIL-S-12514 shall be mounted in a drip-tight enclosure except that a toggle switch shall be provided instead of a push-button station. The switch shall provide for the selection of ON and OFF. The ambient temperature reference shall be 50 degrees centigrade (C). Fuses are not acceptable in lieu of the thermal overload protective device.

3.6.6.6 Power transmission. For type I, II, and III units power shall be transmitted from the gasoline engine to the compressor through a minimum of two B section V-belts, with provision for belt tightness adjustment. All belts shall be replaceable while the unit is mounted on the refrigerated van without removing any major component of the refrigeration unit. V-belt drives shall be in accordance with the Rubber Manufacturers Association Standards for Multiple V-Belt Drives. The condenser and evaporator fans shall be driven with V-belt drive from the compressor or electric motor for type II units with belt tension adjustment by means of an adjustable idler pulley with a minimum diameter of 5 inches mounted between the compressor pulley and the fan drive pulley on the slack side of the belt. The idler pulley bearing shall be of the pressure lubricated type. The type and size shall be that recommended by the bearing manufacturer for this particular application. For type II units the prime mover not being used shall have provisions to "free wheel" and may serve as an idler.

3.6.6.7 Toggle and control switches. All toggle and control switches shall conform to MS 25068 or MS 25098.

3.6.6.8 Battery terminals. Three-eights of an inch diameter brass lug and nut terminals to receive the battery cables shall be located beneath the engine compartment in a location protected from rain or snow. When facing the condenser air inlet, the battery terminals shall be located in the lower right hand portion of the unit and shall be marked for polarity.

3.6.6.9 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum disturbance to adjacent parts of the end item without the use of special tools. Access panels, covers, and plates shall be provided as required for component adjustment, repair, or replacement and shall be fastened with quick disconnect fasteners or with screws. Clearance shall be provided for the following:

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(a) Carburetor adjustments
(b) Removal of the fuel filter
(c) Connecting an auxiliary fuel line
(d) Servicing the air cleaner
(e) Operation of the air cleaner winterization selector lever
(f) Removal and insertion of the oil gage rod and for adding oil with or without a removable spout or funnel
(g) Removal and replacement of the oil drain plug and the drainage of oil. The oil shall drain completely and shall not flow over any part of the end item. An extension to the crankcase drain system is permissible.
(h) Removal and replacement of spark plugs, high-tension cables, breaker points, and capacitors, and for adjustment of the breaker points
(i) Viewing the ignition timing mark

* 3.6.7 Fungus control of coils and electric motors. Starter coils and solenoid coils except potted coils shall be treated in accordance with MIL-T-152. Motor coils with a temperature rise of 40 C. or less shall be coated with fungus resistant varnish conforming to MIL-V-173. Motor coils with a temperature rise exceeding 40 C. shall be given two coats of varnish conforming to type AN, class 105 of MIL-I-24092. The coats shall be applied by the vacuum-pressure, immersion, centrifugal, pulsating pressure, or build-up, method to fill all interstices in the coils and preclude entrapped air or moisture. The sealer coat may also be applied by brushing or spraying.

3.6.8 Finish.

3.6.8.1 Surface preparation. The exterior and interior surfaces of the framework enclosure, base and power unit except where not desirable for proper operation, shall be cleaned and conditioned. Rough or sharp edges and corners, burrs, and slivers shall be removed. Ferrous surfaces on any components to be painted shall first be cleaned and conditioned using the compound specified in 3.4.4.1. Wood surfaces shall be unfinished after being preservative-coated. Aluminum surfaces shall be cleaned and treated by conditioner specified in 3.4.4.2.

3.6.8.2 Priming paint. After surfaces are thoroughly dry from cleaning and conditioning, a coat of primer shall be applied conforming to 3.4.5.1.

3.6.8.3 Finish coat. After primer has thoroughly dried, surfaces shall be given a coat of enamel specified in 3.4.5.2. Enamel shall be applied so that there will be no streaks, sags, pinholes, checks, or cracks. Care should be taken not to paint or allow paint to get on the fins of the condenser or the heat shield gasket or any other portion of the unit where paint would be detrimental, or any portion of the unit extending into the insulated structure.

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3.6.9 Charging refrigeration system. Each complete refrigeration unit shall be dried, evacuated, and fully charged with the refrigerant specified in 3.4.6 and oil specified in 3.4.7 necessary to obtain the proper operating requirements. The moisture content of the refrigerant shall be below 25 parts per million as shown on the liquid sight and moisture indicator as determined by the test specified in 4.8.9.

3.6.10 Electromagnetic compatibility. When specified (see 6.2), the refrigeration units shall be designed and equipped for electromagnetic compatibility in accordance with class IIIC of MIL-STD-461 (see 4.8.12).

3.6.11 Lubrication.

3.6.11.1 Lubricating. Lubricants for meeting test, operating, and storage requirements of this specification shall be used to lubricate wearing parts of this equipment. Particular care shall be exercised in lubricating the gasoline engine and accessories and the evaporator and condenser fans. All points requiring lubrication shall be lubricated before testing. Units run over 50 hours shall be relubricated before shipping.

* 3.6.11.2 Lubrication fittings. Unless otherwise specified (see 6.2), all bearing surfaces shall be equipped with pressure lubrication fittings, conforming to type I of MIL-F-3541, and shall be located to be accessible for greasing with a push operated gun from the front of the units. Each fitting shall be enameled bright red.

3.6.11.3 Tagging lubrication fittings. After lubrication, each item of equipment shall have the fitting or location tagged to indicate the temperature range and grade of lubricant used and recommended frequency of lubrication.

3.6.11.4 Lubrication precaution marking. Where the use of high pressure lubrication equipment, for example 1,000 psi or higher, will damage grease seals or other parts, a suitable warning plate to this effect shall be located at the appropriate place or places. Any other lubrication precautions should likewise be included on a warning plate.

3.6.11.5 Lubrication instructions. Lubrication instructions for performing periodic preventive maintenance as specified (see 6.2) shall be affixed to each refrigeration unit in a visible location.

3.7 Identification.

3.7.1 Identification plate. The identification plate shall conform to type III, composition A, class 1 or 2 of MIL-P-514 and shall be attached by screws or rivets. The refrigeration units shall be marked in accordance with MIL-STD-130 and with the following:

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(a) Nomenclature
(b) Type of unit, capacity of unit, and type of refrigerant and charge
(c) Federal stock number
(d) Contract and technical manual number
(e) Military model number
(f) Manufacturer's name, model, and serial number
(g) Manufacturer's address
(h) U.S.

The plate shall be installed as shown on Drawing 5-13-1957.

3.8 Instruction plates. Instruction plates, made of nonferrous metal, shall be attached to each refrigeration unit. The instruction plates shall have the wiring diagram, refrigeration cycle diagram, and an arrow indicating the direction of the rotation of the motor. The plates shall contain instructions for startup, operating, defrosting, shutdown, and preventive maintenance procedures. Lettering shall not be less than 1/16 inch high.

* 3.9 Control and gage nameplates. All control switches, gages, temperature control valves, and pilot lights shall be identified with nameplates of nonferrous metal, with nomenclature etched on the nameplate. All other valves will be identified by number with a metal tag attached to the valve.

3.10 Workmanship. Sheet metal or structural sections used in fabrication of the refrigeration unit shall not be kinked, or bent. Rivets, when used, shall be driven to completely fill the holes with full sized heads finished and in full contact with the surfaces of the member. Bolts, nuts and screws shall be drawn tight and so secured that they will not work loose under vibration.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3)
b. Initial production inspection (see 4.4)
c. Inspection comparison (see 4.5)
d. Quality conformance inspection (see 4.6)

* 4.3 First article inspection. When a first article is required, the refrigeration unit shall be examined for the defects listed in table I, dimensions specified, and tested as specified in table III. The presence of any defect or failure to meet any test requirement shall be cause for rejection of the first article.

* 4.4 Initial production sample inspection. When an initial production sample is required, the refrigeration unit shall be examined for the defects listed in table I, dimensions specified, and tested as specified in table III. The presence of any defect or failure to meet any test requirement shall be cause for rejection of the initial production sample.

4.5 Comparison sample inspection. When a comparison sample is required, the refrigeration unit shall be examined for the defects listed in table I, dimensions specified, and tested as specified in table III. The presence of any defect or failure to meet any test requirement shall be cause for rejection of the inspection comparison sample.

* 4.6 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.6.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.6.1.1 Gasoline engine test. The gasoline engine (see 3.6.6.1) shall be tested in accordance with test 4.8.17.

* 4.6.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, corrections shall be made to process and items affected.

(a) Wood is immersed in preservative as specified in 3.6
(b) Insulation is installed as specified in 3.6.3.2
(c) Welding and soldering is as specified in 3.6.1, 3.6.2, respectively
(d) Cleaning and conditioning is in conformance with 3.6.8.1
(e) Primer is as specified in 3.6.8.2

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(f) Framework is as specified in 3.6.3.1
(g) Heat shield is as specified in 3.6.3.2
(h) Disassembly of components before soldering and brazing in 3.6.5.16
(i) Electrical equipment, wiring, and accessories are treated as specified in 3.6.7
(j) Copper-8-quinolinolate is added to preservative as specified in 3.4.1.3

* 4.6.3 End item inspection. The inspection shall be all refrigeration units of the same type offered for inspection at one time. The sample unit shall be one completely fabricated refrigeration unit.

* 4.6.3.1 Visual examination. The refrigeration unit shall be examined for the defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped, or area of no film	X	
	Color of finish not as specified		X
	Orange peel, color separation, smudge, fish scale, scratched, strained, overspray, sag, imprint, foreign matter imbedded or finish is otherwise impaired		X
	Rough or sharp edges and corners, burrs, or slivers	X	
Design	Any characteristics not in accordance with the specified requirements	X	
Construction and workmanship general (applicable to all components and assemblies)	Any specified component missing or not specified type (unless otherwise classified herein)	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, decayed or otherwise impaired	X	
Soldering (when applicable)	Missing, not adherent or incomplete	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding and brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured or loose but not affecting operation		X
Rivets	Missing, broken, loose, not peened	X	
Assembly	Any component not properly assembled or secured for example tubing creased at bend restricting or partially restricting interior flow	X	
Switches	Not marked as specified		X
	Not type specified	X	
Marking for identification	Missing, incomplete, not legible, not specified type or size, not proper location, not affixed in the prescribed manner		X
Lubrication instructions	Missing or illegible	X	
Lubrication precautionary marking	Missing, not as specified, not in proper location		X
Instruction plate and installation instructions	Missing, not as specified, not in specified location, illegible or incomplete	X	

* 4.6.3.2 Dimensional, capacity and weight examination. The refrigeration unit shall be examined for compliance with dimensional, capacity, and weight specified. Any dimension, capacity, and weight that does not meet the specified requirements shall be classified a defect. The inspection level shall be S-1 with an AQL of 4.0 expressed in terms of defects per hundred units.

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* 4.6.3.3 End item testing. The refrigeration units shall be tested as specified in table III. The test in 4.8.2 shall be performed on one production unit, selected at random, out of every lot of 25 units or one unit from the lot produced if less than 25, failure to pass this test shall be cause for rejection of the lot. The tests specified in table III for the end item one unit testing, shall be performed on one production unit (see 6.8). Failure of any test shall be cause for rejection of the inspection lot.

* 4.6.4 Examination of preparation for delivery requirements. An examination shall be made to determine that the preservation, packaging, packing and marking complies with the section 5 requirements. Defects shall be scored as specified in table II. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers in the inspection lot. Examination for defects in preservation and packaging shall be in accordance with MIL-P-116.

TABLE II. Examination of preparation for delivery requirements

Examine	Defect
Markings (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Materials	Any components missing, damaged, or not as specified

4.6.4.1 Examination of shipping containers. Examination for defects in fastening devices, construction, blocking, bracing, anchoring, closure, and strapping shall be in accordance with MIL-C-104.

* 4.7 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

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4.8 Tests.

TABLE III. Test schedule

Test	First article	Initial production	Comparison test	End item each lot	End item each unit	End item one unit	Test paragraph	Requirement para.
Operation	X	X	X	-	X	-	4.8.1	3.5 and 3.6.5.9
72 hr.	X	X	-	X	-	-	4.8.2	3.5
0 F.	X	-	X	-	-	X	4.8.3.2	3.5.1.1.1
+ 35 F.	X	-	X	-	-	X	4.8.3.3	3.5.1.1.2
*Heating	X	-	X	-	-	X	4.8.3.4	3.5.1.1.3
High temp.	X	-	X	-	-	X	4.8.4	3.5
Defrost	X	X	X	-	-	X	4.8.5	3.6.5.6
Rainfall	X	X	-	-	-	X	4.8.6	3.5
Impact	X	-	-	-	-	X	4.8.7	3.5
Shock	X	-	-	-	-	X	4.8.8	3.5
Ref. moisture	X	X	-	-	X	-	4.8.9	3.6.9
Ref. leak	X	X	-	-	X	-	4.8.10	3.6
Lifting ring	X	-	-	-	-	X	4.8.11	3.6.3.1
Electromagnetic	X	-	-	-	-	X	4.8.12	3.6.10
Storage	X	X	-	-	-	X	4.8.13	3.5
Road endurance	-	X	X	-	-	-	4.8.14	3.5
Pull down	-	X	X	-	-	-	4.8.15	3.5
Sand and dust	-	X	X	-	-	-	4.8.16	3.5
Engine test	-	X	-	X	-	-	4.8.17	3.6.6.1

* 4.8.1 4-hour operating test. The refrigeration unit shall be mounted to an inclosure. All refrigeration unit components shall be in place, and the refrigeration unit doors shall be in a closed position. The refrigeration unit shall be operated for 4 hours to determine that all mechanical equipment is functioning as intended in accordance with section 3. The unit shall be stopped and immediately restarted every 30 minutes during operating period. The by-pass valve, thermometer, crankcase pressure regulator, and head pressure regulator valve shall be checked for proper operation, setting and calibration. Prior to completion of the 4-hour test, the airflow to the condenser shall be blocked off and the unit operated to check compressor discharge pressure safety switch and red warning light to determine compliance with 3.6.5.9. Malfunction of mechanical equipment or improper operation of any controls and safety switch shall constitute failure of this test.

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4.8.2 72-hour operating test. A 72-hour test shall be conducted as follows:

(a) The refrigeration unit shall be fastened to the evaporator opening of a calorimeter or other insulated structure.

(b) A total heat load of 7000 Btuh plus or minus 1000 Btuh shall be imposed on the unit.

(c) The refrigeration unit shall be operated 36 hours under rating conditions of 3.5.1.1.1 and for 36 hours under the rating conditions of 3.5.1.1.2. During this test, suction and discharge pressures and ambient and inclosure temperatures shall be recorded on automatic recording instruments and upon completion of the test this data shall be made available to the contracting officer. During the test the unit shall be stopped every 4 hours, the amount of oil shall be checked, oil added if necessary, all data shall be recorded, and the unit shall be restarted within 10 minutes. Normal maintenance shall be performed during this test. Failure of the unit to operate or failure to maintain rated conditions for 72 hours shall be cause for failure of this test. Engines which fail during this test will be repaired or replaced and the test rerun.

4.8.3 Capacity tests.

4.8.3.1 Calorimeter calibration. The calorimeter calibration shall be run once prior to capacity tests to establish the coefficient of heat leakage for the insulated calorimeter, to be used for capacity calculation. The coefficient of heat leakage shall not exceed 25 Btuh per degree F. and the volume shall not be less than 400 cubic feet. A refrigeration unit shall be bolted to the evaporator opening in the calorimeter with a minimum 5 foot clearance on the front of the unit and the overall heat leakage coefficient shall be determined as follows:

(a) An electric fan of capacity to insure that no uncirculated air pocket will exist, shall be installed inside the calorimeter. The electrical energy of the fan motor shall be measured by the same meter used for heating elements in subparagraph (c) and this heat equivalent shall be added to other sources of heat added to the calorimeter.

(b) Four thermocouples, shielded from the effects of radiant heat, shall be installed inside the calorimeter and positioned within 6 inches of the ceiling and floor at two opposite corners. The interior temperature shall be taken as the average of the four thermocouples.

(c) Heating elements, which may be incandescent lamp bulbs coupled two in series to cut down on the voltage to minimize radiation, shall be installed approximately in the center of the calorimeter. Electrical heat input shall be measured by a watt-hour meter.

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(d) Eight thermometers or thermocouples, shielded from the effects of the radiant heat, shall be installed approximately 18 inches from each corner in the room in which the calorimeter is placed. The ambient temperature shall be taken as the average of these eight temperatures.

(e) All openings in the calorimeter shall be sealed and the calorimeter shall remain undisturbed throughout the test.

(f) The ambient temperature shall then be maintained constant within plus or minus 2 F. Sufficient steady, nonvarying heat shall be added to the interior of the calorimeter to maintain a minimum of 60 F. and a maximum of 100 F. temperature difference between the interior and exterior of the calorimeter.

(g) The test shall be considered concluded when eight consecutive one-half hourly readings indicate that:

1. The ambient temperature has been maintained constant within plus or minus 2 F.

2. The temperature inside the calorimeter has been maintained constant within plus or minus 1 F.

(h) The overall heat leakage coefficient U is given by:

$$U = \frac{H_1}{(T_1 - T_2)}$$

U = Calorimeter heat leakage factor

H_1 = Sum of all electrical heat energies added to the structure Btuh

T_1 = Average interior temperature of insulated structure (100 F. to 155 F.)

T_2 = Average ambient temperature, $T_1 - T_2$, shall be in the range of 60 F. to 100 F.

(i) Failure to meet the heat leakage coefficient of 25 Btuh per degree F. shall be cause to reject the test calorimeter.

4.8.3.2 0 F. refrigerator test. The capacity of the refrigeration unit for low temperature shall be determined by the method described below to determine compliance with 3.5.1.1.1. Inability of the unit to meet the requirements shall constitute failure of the test. Recorded data shall be required and shall include

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all specified temperature readings, electric heat input, suction and discharge pressures and complete calculations. All panels and doors of the refrigeration unit shall be in the normally closed position.

(a) All accessories and appurtenances shall be in their proper place.

(b) Thermocouples, shielded from the effects of radiant heat, shall be installed in the air streams entering the evaporator and leaving the evaporator fan.

(c) Shielded thermocouples shall be located at the approximate center of four equal areas of the condenser face. The condenser inlet temperature will be taken as the average of these four temperatures.

(d) Recording pressure gages shall be used for the compressor suction and discharge pressures.

(e) The procedures described in 4.8.3.1 (b), (c), (d), and (e) shall be followed for this test.

(f) The refrigeration unit shall then be operated with air entering the condenser at 110 F., plus or minus 2 F., and sufficient steady, nonvarying heat shall be added to the interior of the calorimeter to maintain 0 F. temperature of air entering the evaporator, plus or minus 1 F. A watt-hour meter shall be used to indicate the quantity of electrical energy added. Air velocity shall not exceed 100 feet per minute (ft/min) at any point three feet from the refrigeration unit, excluding the evaporator area.

(g) The test shall be considered concluded when eight consecutive one-half hourly readings indicated that:

1. The average calorimeter ambient temperature, T_3, has been maintained at steady state of plus or minus 2 F. at a temperature of 100 to 110 F.

2. Return air temperature, T_4, entering the evaporator has been maintained at 0 F., plus or minus 1 F.

3. The net refrigerator capacity shall be considered as the sum of the electrical energy, in Btuh, added during this test plus the heat leakage gain in Btuh, as calculated from the calibration specified in 4.8.3.1.

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The net refrigeration capacity is given by:

$$Q = U (T_3 - T_4) + H_2$$

U = Overall heat leakage coefficient Btu as determined by paragraph (h) in 4.8.3.1

H_2 = Sum of all heat energy added to the interior of the calibrated box in Btuh ((f) above)

4.8.3.3 35 F. refrigeration test. The procedure specified in 4.8.3.2 shall be followed with the following exceptions to determine compliance with 3.5.1.1.2.

(a) In paragraph 4.8.3.2 (f), sufficient heat shall be added to maintain 35 F. temperature, plus or minus 1 F., of air entering evaporator in place of 0 F. temperature specified.

(b) In paragraph 4.8.3.2 (g) 2, temperature of the air entering evaporator has been maintained at 35 F. plus or minus 1 F. in place of temperature specified. Inability to meet the requirements shall constitute failure of the test.

4.8.3.4 Heating test, type III unit. The heating capacity of the refrigeration unit shall be determined by repeating all procedures specified in 4.8.3.2 with the following exceptions to determine compliance with 3.5.1.1.3.

(a) Paragraph (f) - the refrigeration unit shall be operated on a heating cycle in an ambient of minus 25 F. or lower. Sufficient cooling shall be added to the insulated calorimeter to maintain a minimum temperature of 35 F. plus 2 F., minus 0 F.

(b) Paragraph (g) - the test shall be considered concluded when four subsequent consecutive 30 minute readings indicate that:

1. The insulated calorimeter temperature had been maintained in an ambient temperature of minus 25 F. or lower (T_5).

2. The temperature of the air entering the evaporator has been maintained at 35 F. plus 2 F., minus 6 F. (T_6).

(c) Paragraph (h) - the net heating capacity "Q" in Btuh will be calculated by:

$$Q = U (T_6 - T_5) + H_3$$

Where H_3 = Sum of all heat energy removed from interior of the calibrated structures. Failure of the unit to meet the requirements shall constitute a failure of this test.

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4.8.4. High temperature test. With the unit remaining in position after completion of the capacity tests, the temperature of air entering the condenser shall be increased to a minimum of 120 F. to determine compliance with 3.5. The unit shall then be started and operated in this ambient for 4 hours with its control set for 0 F. or below. Air velocity shall not exceed 100 ft/min at any point 3 feet from the refrigeration unit excluding the evaporator area. Inability of the unit to start and operate for 4 hours at 120 F. shall constitute failure of this test.

4.8.5 Defrost test. With the unit operating and the insulated structure maintained at 0 F. plus and minus 10 F. the evaporator will be frosted up with approximately 20 pounds of water on the coil surface. After the coil is heavily frosted, the unit will be set on the defrost cycle, and the defrost condensate will be collected outside the refrigerated compartment and weighed for compliance with 3.6.5.6.1 and 3.6.5.6.2, as applicable. Observation shall be made to determine that the coil is free of frost when the red light is on. Inability of the unit to defrost as specified including overloading shall constitute failure of the test.

4.8.6 Rainfall test. The refrigeration unit shall be installed in operational position and subjected to a simulated rainfall to determine compliance with the rainfall requirements of 3.5. Fresh water shall be sprayed on the unit at a rate of 3 inches per hour at an average downgrade angle of 45 degrees to simulate rainfall against the exposed sides of the unit. The unit shall be exposed to the spray for 30 minutes while the unit is not operating and then shall be started and operated for 30 minutes while the spray continues. Inability of the unit to start and continue operating shall constitute failure of the test.

* 4.8.7 Impact test. The refrigeration unit shall be mounted and operated on a test frame constructed to simulate mounting on a semitrailer. The test frame shall be raised and dropped so that the unit receives an impact of 5G magnitude. This impact shall be repeated for a total of 200,000 times; 50,000 times with the unit not running, followed by 100,000 times with the unit running, and 50,000 times with the unit not running, to determine compliance with 3.5. Every 4 hours the test shall be stopped and the unit visibly examined for defects. Minor tightening of loose nuts may be made; bolts may be tightened and a record of all corrections shall be made available to the contracting officer. After completion of the test the unit shall be visibly examined for defects such as deformation, loose components, strain cracks in the frame, ruptured lines and leaks. After passing the visible examination the unit shall be started and run. The unit shall be examined for any mechanical malfunction and leak tested. Any deformation, malfunction, cracks in frame, refrigerant leaks, or failure to operate shall constitute failure of this test. Unit shall be serviced with gas and oil before restarting.

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4.8.8 Shock test. The complete refrigeration unit serviced and ready for operation and mounted on a drop frame constructed as shown on Drawing 5-13-1957, shall be shock tested as follows to conform with the requirements of 3.5:

(a) The unit shall be mounted on the drop frame in the same manner the unit would be mounted on a trailer body. The drop frame shall be lifted vertically and free dropped on a concrete floor from a height of 6 inches. The drop shall be repeated nine times. Upon completion of this test the unit shall be examined and the following shall constitute failure of this test:

1. Permanent distortion of 1/32 of an inch on the frame, measured from center punch points located on both sides of the frame, one in the upper corner and one on the diagonally opposite lower corner.
2. Any sag in the evaporator section greater than 1/32 of an inch, measured by scribing lines on the evaporator section parallel to the drop frame so that movement relative to the frame can be measured.
3. Any visible change in the alignment of the condenser or evaporator fan and shroud, greater than 1/8 of an inch.
4. Any refrigerant leaks, other observable damage, movement of the components, misalignment of components or unit.

(b) After successfully passing (a) the unit still mounted in the drop frame shall be started and operated for 30 minutes and the unit leak tested again. Any leaks, operational malfunctioning of components, abnormal vibration, or other irregular operation shall constitute failure of this test.

4.8.9 Refrigerant moisture test. The refrigeration unit shall be operated for at least one hour. The moisture indicator shall then be observed to determine compliance with 3.6.9 as shown by the component manufacturer's color chart date. Moisture exceeding the requirements shall constitute failure of this test.

4.8.10 Refrigeration leakage test. Prior to applying lacquer specified in 3.6.5.16 the refrigeration leakage test shall be performed to determine compliance with 3.6. The test for leakage shall be made with a halide leak detector or electronic leak detector. Inability to meet the leakage requirements shall constitute failure of this test.

* 4.8.11 Lifting ring test. The refrigeration unit shall be lifted free so that each lifting ring supports one-half of the weight. It shall be observed that the refrigeration unit remains in a vertical position not more than 2-inches out of plumb over the length of the unit, in compliance with the lifting ring locations requirements of 3.6.3.1. To test for strength of each lifting ring, the refrigeration unit shall be lifted free by each lifting ring singularly. Failure of the lifting ring to meet the requirements of 3.6.3.1 shall constitute failure of this test.

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* 4.8.12 Electromagnetic compatibility test. When electromagnetic compatibility is required the refrigeration unit shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

* 4.8.13 Storage test. The completed refrigeration unit shall be subjected to a temperature of plus 155 F. and 10 to 20 percent humidity for 4 hours and then to a temperature of minus 30 F. for 6 hours to determine conformance to the requirements of 3.5. Any nonconformance with 3.5 shall constitute failure of this test.

4.8.14 Road endurance test. The assemblage shall be capable of withstanding 3600 miles of road operation and mileage as follows to determine compliance with 3.5:

(1) The road mileage schedule shall be as follows:

(a) Hard surfaced, paved, 1200 miles up to 50 miles per hour.
(b) Level cross-country, 1200 miles up to 25 miles per hour.
(c) Hilly cross-country, 1140 miles up to 25 miles per hour.
(d) Belgian block, 60 miles up to 20 miles per hour.
(e) The speed of the vehicle shall be varied up to the specified speed throughout each segment of the test.

(2) The above mileage shall be accomplished over all courses with the trailer unloaded. The refrigeration unit shall be running, with control set at 0 F. for 10 hours. The unit shall be stopped and the gasoline tank shall be filled, the oil shall be checked and oil added when necessary, the entire refrigeration unit shall be checked for loose bolts, nuts, screws, or other parts and any minor repairs shall be made. At the end of 1/2 hour the refrigeration unit shall be restarted and the road test continued. During the conduct of the road test all scheduled maintenance will be performed on the refrigeration unit, and the semitrailer. Inability of the unit to operate for 3600 miles as specified shall constitute failure of this test.

* 4.8.15 Pull down test. To determine compliance with 3.5 the refrigeration unit attached to an empty refrigerator semitrailer (series M349). The assemblage shall be heat soaked for 4 hours in an ambient of 110 F. While the ambient temperature is maintained at 110 F., air inlet temperature is at 110 F., the unit shall be started and pulled down to 0 F. in 8 hours or less. Records shall be made every 30 minutes of the van temperature, head pressure, suction pressure, and air ambient temperature. Inability of the unit to pull down to 0 F. within 8 hours shall constitute failure of this test.

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* 4.8.16 Sand and dust test. The refrigeration unit shall be mounted in a van and operated for 2 hours while subjected to blowing sand and blowing dust as specified in AR 70-38, to determine compliance with 3.5. During the exposure, the gages on the unit will be observed for improper operation and the unit will be observed for any excessive noise and vibration. The inside of the van will be examined for any penetration of sand or dust through the refrigeration unit. The unit shall be run an additional 500 hours to determine if the dust or sand has penetrated bearings or other moving parts. While the unit is undergoing the 500 hours duration portion all other initial production tests should be conducted. Inability of the unit to operate or any malfunction, caused by sand or dust, shall constitute failure of this test.

4.8.17 Gasoline engine control test. The first produced gasoline engine shall be examined and tested in accordance with test method 4000 of MIL-STD-1400. Thereafter one gasoline engine selected at random out of 50 shall be tested. When less than 50 gasoline engines are submitted one gasoline engine shall be tested as above.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation and packaging. All exposed, unpainted, uncoated ferrous metal surfaces shall be cleaned process C-1 of MIL-P-116 and thoroughly dried. All non-contacting surfaces cleaned as specified herein (surfaces that do not contact other surfaces in operation) shall be coated with type P-1 preservative; contacting surfaces cleaned as specified herein (surfaces that contact other surfaces in operation) shall be coated with type P-6 preservative in accordance with MIL-P-116. Under no circumstances shall preservative of any kind be applied to the evaporator section including the fan.

* 5.1.1.1.1 Refrigeration system. The refrigerant charge shall be pumped down and the appropriate valves, including the receiver outlet valve and compressor suction and discharge valves, closed and capped. Closed valves shall be tagged or otherwise clearly marked with instructions for restoring to service. Warning tags shall be prominently affixed to the controls, and shall caution against operating the unit without opening valves.

5.1.1.1.2 Engine. The gasoline engine shall be prepared in accordance with the type II, method I, level A requirements of MIL-E-10062.

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5.1.1.1.3 Pulleys. Belts shall be removed and the exposed, uncoated ferrous metal surface of shafts and pulleys shall be coated with compound conforming to TT-P-664. After coating has dried, belts shall be replaced and relieved of any tension.

5.1.1.1.4 Protection of preservative. Surfaces from which preservative compound might be removed by contact with packaging and packing materials shall be covered or wrapped with greaseproof barrier-material conforming to type I or II, grade A, class 1 or 2 of MIL-B-121.

5.1.2 Level C (commercial packaging). Each refrigeration unit shall be preserved and packaged to afford adequate protection against physical damage and corrosion during shipment from the supplier to the first receiving activity. The preservation and packaging shall be the same as that normally used by the supplier for retail distribution.

5.2. Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each refrigeration unit, preserved and packaged as specified in 5.1, shall be packed in the operating position within a crate conforming to type II, class 1 or 2, style a of MIL-C-104. The contents of each crate shall be anchored, blocked and braced in accordance with the appendix of MIL-C-104.

* 5.2.2 Level C (commercial packing). Refrigeration units, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking requirement by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Precautionary and handling markings shall be applied to each shipping container.

6. NOTES

6.1 Intended use. The refrigeration units covered by this specification are intended for worldwide Military field service under severe operational and load conditions, directly exposed to the elements except for the evaporator section. The units are used with M349 series semi-trailer refrigerators covered by MIL-S-62034 for the preservation of temperature sensitive products.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of refrigeration unit to be furnished (see 1.2).

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(c) First article. When a first article is required (see 3.1, 4.3, and 6.3).
(d) When initial production testing sample is required (see 3.1.1, 4.4, and 6.4).
(e) When inspection comparison testing sample is required (see 3.1.2, 4.3.5, and 6.5).
(f) When electromagnetic compatibility is required (see 3.6.10).
(g) When lubrication fittings are other than specified (see 3.6.11.2).
(h) Lubrication instructions (see 3.6.11.5).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

* 6.5 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

* 6.6 Where installation or usage of refrigeration unit is not expected immediately upon arrival, level C preservation and packaging should not be used. If experience indicates deterioration during period between receipt and usage, use level A preservation and packaging.

6.7 Contract data requirements. Any requirements for instruction manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.8 End item tests. Performance of the one unit end item tests as specified in table II should be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

6.9 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the

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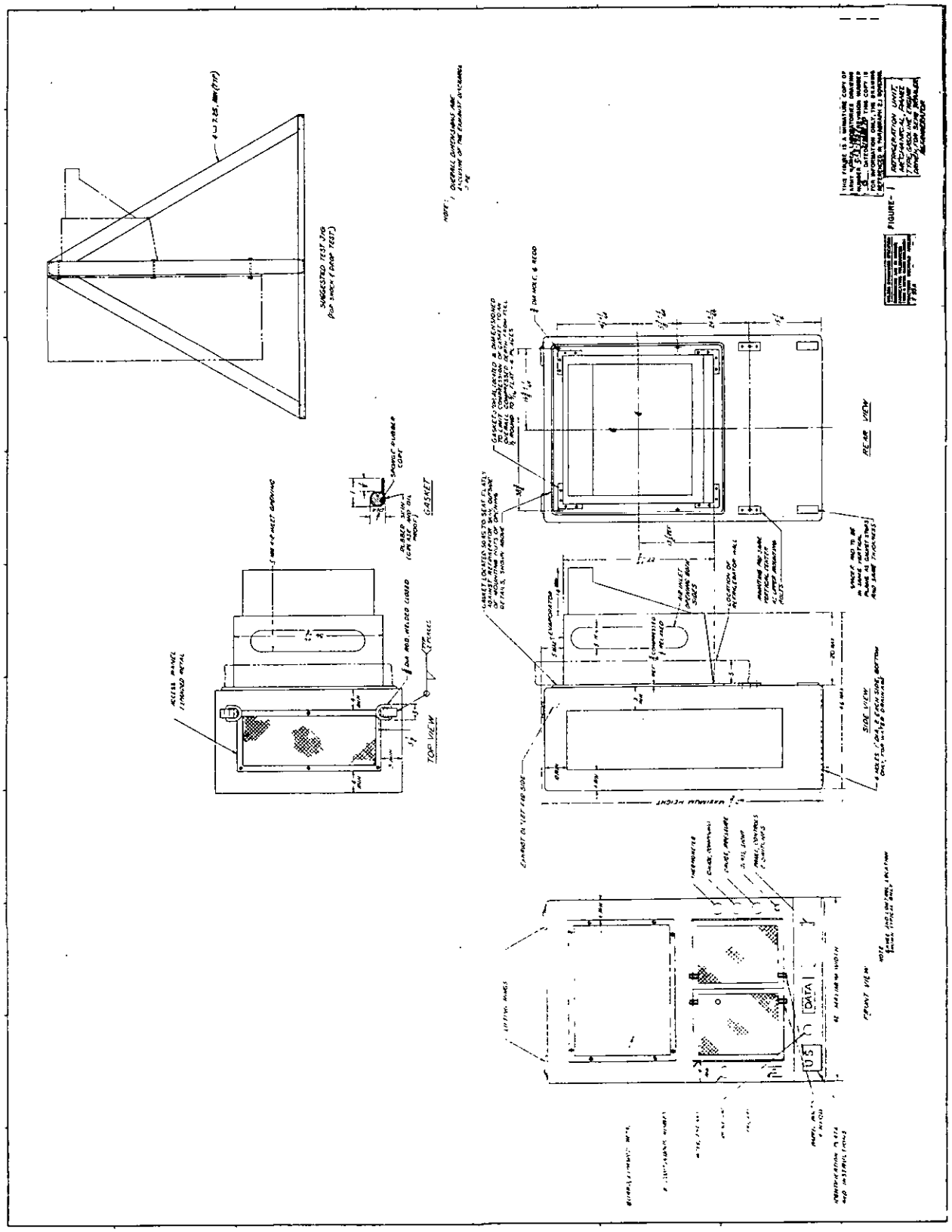
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DOCUMENT IDENTIFIER AND TITLE MIL-R-10735K - REFRIGERATION UNITS, MECHANICAL, PANEL TYPE, GASOLINE ENGINE AND ELECTRIC MOTOR DRIVEN, FOR SEMITRAILER REFRIGERATORS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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